When Does Data Quality Help Human Deferral? Governance-Constrained Review Allocation in Knowledge-Based Decision Systems

Yeshwanth Reddy Kunam[a,*]

[a] Doctor of Business Administration, Midwest University, 851 Parr Road, Wentzville, MO 63385, United States

* Corresponding author. E-mail: yeshwanthkunam05@gmail.com

Revised author-review draft (October 2026). Not yet submitted or peer reviewed.

Highlights

- Exact sorting solves governance-constrained review allocation in both regimes.
- Backlog depends on staffing, not on how mandatory cases are prioritised.
- Learned deferral beat cost-aware heuristics only with thousands of review logs.
- Quality inputs helped most when missingness was informative about outcomes.
- Quality-triggered rules missed three-quarters of records with invisible defects.

Abstract

Organisations that automate decisions must decide which records a model may decide alone, where scarce human review reduces expected loss, and whether staff can absorb the review demand that governance rules make compulsory. We study these choices in a knowledge-based routing system combining a calibrated predictor, a data-quality rule base, a two-head loss estimator that learns from incomplete review logs, and an allocator that enforces review capacity and holds mandatory cases it cannot serve. Sorting solves this governance-constrained allocation exactly in feasible and overloaded regimes, and in a multi-period system the backlog does not depend on how mandatory cases are prioritised. We compare the system with published learning-to-defer surrogates, differential triage and cost-aware heuristics under matched information, rules, costs and capacity. The evaluation spans four credit datasets (one with natural missingness), nine data-quality conditions including defects invisible to validation checks, and five simulated reviewer models, over 20 corrected repeated splits. Pooled, the proposed router was the best learning-based method but 1.2% costlier than Chow's cost-risk rule. Learned routing overtook that rule only with roughly 1,800 or more training records, and with 6,000 it beat every comparator on all splits.

Quality information reduced cost mainly when missingness was informative about outcomes (1.2–3.5%), by improving estimates of automated rather than reviewer error. In a 40-day simulation with a deterioration episode, staffing determined backlog and service completion far more than routing did. Quality-triggered rules blocked most visibly degraded records but missed three-quarters of records with invisible defects.

Keywords: Knowledge-based systems; Learning to defer; Data quality; AI governance; Human–AI collaboration; Review capacity; Managerial decision-making

1. Introduction

Organizations that automate operational decisions with machine-learning models rarely automate everything. Credit, claims, compliance and service-eligibility decisions are typically routed through a hybrid process in which a model decides routine records and human reviewers examine the remainder. AI-governance frameworks describe this division of labour in terms of accountability, data controls and human oversight [1–4], and recent regulation makes human oversight an explicit obligation for high-risk uses such as creditworthiness assessment [5]. Requiring oversight, however, does not establish that oversight is effective [6]. Managers still have to decide which records may be automated, where scarce review effort reduces expected loss, and whether staffing can absorb the review demand that governance rules make compulsory.

Two research streams address parts of this problem. Learning to reject and learning to defer (L2D) decide when a model should abstain or pass a case to a human, trading model error against the cost or error of deferral [7–12]. Algorithmic triage estimates human and model error separately and automates the cases where the model is relatively stronger [13, 14], and recent work adds expert workload constraints [15] or limited expert predictions [16]. In parallel, data-management research treats completeness, validity and timeliness as measurable properties of the inputs a model receives [17–19]. Governance practice links the two: input-validation rules and model-risk controls often require that records failing quality checks be reviewed rather than automated [20].

What is less well understood is how these elements interact when data quality deteriorates. Mandatory-review rules triggered by quality checks increase review demand precisely when review capacity is fixed; records that cannot be reviewed are neither automated nor resolved; and a quality signal that is informative about model error may or may not be informative about reviewer error. Existing deferral studies typically evaluate accuracy–coverage trade-offs with no compulsory review and no unresolved cases, and data-quality studies rarely evaluate the downstream allocation of human effort. An earlier draft of the present study proposed a compact quality-conditioned gain model with two quality rules and reported that, under a reviewer whose errors were simulated from the same quality signals given to the router, the quality-aware policy changed expected cost by only about 0.2% relative to a feature-aware comparator, with an interval that crossed zero. That result raised three questions this revision is designed to answer: whether the proposed method adds anything beyond published deferral algorithms, whether its apparent value depended on a reviewer simulation that favoured its own representation, and under which conditions observable quality information is useful at all.

This paper therefore asks:

RQ. Under what data-quality and review-capacity conditions does quality-aware deferral improve decision cost and service completion while satisfying explicit governance rules?

We study the question with a knowledge-based routing system consisting of a calibrated predictor, a two-rule quality knowledge base, a two-head loss estimator and an exact capacity-constrained allocator. The contributions are as follows.

1. **A governance-constrained allocation problem and its exact solution.** We formalise routing with automation, review and hold actions under a review budget, mandatory-review rules and a service-first constraint. We show that sorting solves the estimated problem exactly in both the feasible and the overloaded regime (Proposition 1). The overloaded case corrects the earlier draft, which ranked mandatory records by estimated gain even though automation loss is irrelevant for records that cannot be automated. We also show that, in a multi-period system, backlog size, workload and waiting cost do not depend on how mandatory records are prioritised (Proposition 2), which separates staffing decisions from prioritisation decisions.
2. **A two-head estimator that learns from incomplete review logs.** Expected review benefit is decomposed into an automation-loss head trained on all labelled records and a reviewer-loss head trained only on records with a logged review. This follows the triage logic of Raghu et al. [13] but is tied to the allocation problem above and uses the reviewer-loss head directly for overload ordering.
3. **A matched benchmark against published deferral methods.** We compare the system with the cost-sensitive cross-entropy surrogate of Mozannar and Sontag [10], the one-vs-all surrogate of Verma and Nalisnick [11], differential triage [13] and Chow-type cost-risk rejection [7]. All methods receive the same information, rules, costs and capacity and are trained on noisy, incomplete review logs rather than on expected reviewer losses.
4. **A conditions analysis rather than a superiority claim.** Five reviewer models, four of whose errors do not depend on the router's quality signals and two of which err on information the router cannot observe, are crossed with two families of data defects: defects visible to completeness and range checks, and domain-valid defects (informative missingness, stale records and mis-keyed codes) that such checks largely miss. We add an external dataset with naturally occurring missingness. We report where quality information helps, where it does not and where it hurts, and how much review history a learned router needs before it beats a cost-aware heuristic. We also report managerial outcomes: review workload, backlog, waiting time, service-level attainment, override rate and cost per completed case.

Section 2 reviews related work. Section 3 formulates the problem and the method. Section 4 describes the experimental design and prespecified hypotheses. Section 5 reports results, Section 6 discusses implications, and Section 7 sets out limitations.

2. Related work

2.1. Rejection, deferral and triage

Chow's rule rejects an input when the expected loss of the best decision exceeds a fixed rejection cost [7]; modern variants learn the rejector jointly with the classifier [8, 21] or wrap black-box models [22]. Learning to defer

replaces the constant rejection cost with the loss of a specific human decision maker. Madras et al. [9] introduced the setting; Mozannar and Sontag [10] gave a consistent cost-sensitive surrogate; Verma and Nalisnick [11] proposed a one-vs-all surrogate with calibrated deferral probabilities; and later work studied multiple experts [23, 24], underfitting [25], exact mixed-integer formulations [12] and human–AI dependence [26]. Triage methods estimate human and algorithmic error separately and rank cases by their difference [13], or learn the model and triage policy jointly [14]. Several studies relax the assumption of complete expert data: Hemmer et al. [16] learn with limited expert predictions, and Alves et al. [15] combine a human-error model with a constrained assignment step under analyst workload limits. Budgeted multiple-expert deferral [27], congestion-aware deferral [28] and multi-stage human–AI frameworks [29] extend the setting further. Bansal et al. [30] show that the most accurate model is not necessarily the best teammate once human behaviour is considered.

None of these methods, to our knowledge, combines (i) mandatory review triggered by input-quality rules, (ii) a hard capacity constraint that can be exceeded by mandatory demand, and (iii) explicit accounting for unresolved cases. Table 1 summarises the positioning. Our claim is correspondingly narrow: we do not propose a new consistent surrogate, and we do not claim to be the first quality-aware router [31].

Table 1. Positioning relative to deferral and triage studies. "Partly" denotes cost asymmetry handled by weighting rather than in the objective.

Study	Asymmetric costs	Review capacity	Mandatory rules	Unresolved cases / backlog	Input-quality signals	Incomplete reviewer logs
Chow [7]	yes	no	no	no	no	n/a
Madras et al. [9]; Mozannar & Sontag [10]	yes	no	no	no	no	no
Verma & Nalisnick [11]	no	no	no	no	no	no
Raghu et al. [13]; Okati et al. [14]	partly	coverage level	no	no	no	no
Hemmer et al. [16]	no	no	no	no	no	yes
Alves et al. [15]	yes	per-expert workload	no	no	no	one prediction per case
Farinhas et al. [31]	no	budget	no	no	output-quality estimates	n/a
Saketos et al. [28]	yes	congestion	no	queueing	no	online
This study	yes	batch budget	quality-triggered	holds and multi-period backlog	completeness and range	yes (40% logged, single decision)

2.2. Data quality and validation

Data quality is multidimensional, spanning accuracy, completeness, consistency and timeliness [17]. Production machine-learning pipelines validate schemas, ranges and distributions before scoring [18, 19], and IoT and en-

terprise settings add timeliness and provenance concerns [32]. Two properties of such checks matter for deferral. First, they measure the visible symptoms of defects (an empty field, an implausible value) rather than the defects themselves; stale but plausible values and mis-keyed but valid codes pass them. Second, missingness may be informative about the outcome rather than random [33], so a completeness signal can carry predictive information that has nothing to do with reviewer accuracy. Both properties are central to our experimental design.

2.3. Governance, oversight and managerial decision-making

Governance frameworks emphasise accountable processes, documented controls and human oversight [1–4, 34]. Supervisory guidance on model risk requires validation, monitoring and controls on input data [20], and the EU AI Act requires that high-risk systems be designed so that natural persons can effectively oversee them [5]. Green [6] warns that oversight requirements can legitimise flawed systems if reviewers lack the capacity or information to correct them. From a managerial perspective, human–AI complementarity [35] and organisational AI capability [36] depend on how human effort is deployed. Explainable decision-support work in knowledge-based systems links model outputs to business actions [37] but seldom models the capacity of the humans who act on them. Our audit record, which stores probability, quality measures, rule flags, estimated losses and actions, documents routing decisions; it is not an explanation of model behaviour in the XAI sense [38–40].

3. Problem formulation and method

3.1. Setting and notation

A batch contains n records with feature vectors x_i, which may be degraded, and unknown labels $y_i \in \{0, 1\}$. A base predictor produces a calibrated probability p_i of class 1. With false-negative cost c_{FN} and false-positive cost c_{FP} (5 and 1 in the main analysis), the automated decision is $\widehat{y}_i = \mathbb{1}[p_i \geq c_{FP}/(c_{FP} + c_{FN})]$ and its loss is

$$\ell_i^A = w(y_i)\,\mathbb{1}[\widehat{y}_i \neq y_i], \qquad w(1) = c_{FN},\ w(0) = c_{FP}. \tag{1}$$

A human reviewer errs with probability e_i, so the expected reviewed loss is $\ell_i^H = w(y_i)\,e_i$. A review costs a fee c_R (0.05), and a record that is neither automated nor reviewed is held at penalty c_H (1.5), representing unresolved service rather than a correct decision. The expected benefit of review is

$$g_i = \ell_i^A - \ell_i^H - c_R. \tag{2}$$

3.2. Quality measures and the rule base

The quality monitor computes two observable measures from the record alone: the missing-field fraction u_i and the out-of-range fraction v_i, where a non-missing value is out of range if it falls outside $[Q_{.01,j} - 0.1t_j,\ Q_{.99,j} + 0.1t_j]$ with $t_j = \max(Q_{.99,j} - Q_{.01,j}, 1)$, using quantiles of predictor-training rows. Both fractions use all

features as denominator. The router sees these measures, never the injected-defect labels.

The knowledge base contains two forward rules,

$$\text{R1: } u_i \geq \tau_u \Rightarrow m_i = 1, \qquad \text{R2: } v_i \geq \tau_v \Rightarrow m_i = 1, \tag{3}$$

with $(\tau_u, \tau_v) = (0.5, 0.25)$ in the main analysis. These encode an organisational policy that heavily incomplete or implausible records must not be decided automatically. They are illustrative, and Section 5.7 evaluates stricter, more lenient and absent rule sets. Let $M = \{i : m_i = 1\}$.

3.3. Governance-constrained allocation

Each record receives exactly one action: automate (a_i), review (r_i) or hold (h_i). Given capacity $B = \lfloor bn \rfloor$, the allocation problem is

$$\min_{a,r,h} \ \frac{1}{n} \sum_i \left[a_i \ell_i^A + r_i(\ell_i^H + c_R) + h_i c_H \right] \tag{4}$$

subject to $a_i + r_i + h_i = 1$, $\sum_i r_i \leq B$, $a_i \leq 1 - m_i$ (no automation of mandatory records), $h_i \leq m_i$ (only mandatory records may be held), and the service-first constraint $\sum_i h_i = (|M| - B)^+$, which forbids voluntary holds when mandatory review is feasible. In deployment, ℓ_i^A and ℓ_i^H are replaced by estimates.

Proposition 1 (exact allocation). *(a) If $|M| \leq B$, an optimal solution reviews every mandatory record and the* $\min\{B - |M|, |\{i \notin M : g_i > 0\}|\}$ *optional records with the largest g_i, and automates all other records. (b) If $|M| > B$, every feasible solution reviews exactly B mandatory records and no optional record, and holds $|M| - B$ records; an optimal solution reviews the B mandatory records with the smallest ℓ_i^H. In both regimes the optimal routes do not depend on c_H.*

Proof. (a) The service-first constraint gives $\sum h_i = 0$, so every $i \in M$ has $h_i = a_i = 0$ and $r_i = 1$, using $|M|$ reviews. Each optional record is either automated or reviewed, and reviewing it changes the objective by $-g_i/n$. Minimising (4) is therefore equivalent to choosing at most $B - |M|$ optional records to maximise $\sum g_i$, which is achieved by taking the largest positive gains (an exchange argument). (b) Exactly $|M| - B$ mandatory records are held. The remaining B mandatory records cannot be automated and so must be reviewed, which exhausts capacity. The objective equals $\frac{1}{n}[\sum_{i \notin M} \ell_i^A + (|M| - B)c_H + \sum_{i \in R}(\ell_i^H + c_R)]$ for the reviewed mandatory set R; only the last term depends on the choice, and it is minimised by the B smallest ℓ_i^H. c_H enters only through the constant $(|M| - B)c_H$. □

Proposition 1(b) corrects the earlier draft, which ordered mandatory records by estimated gain $\hat{g}_i$ under overload. Because a mandatory record can never be automated, ℓ_i^A is irrelevant to that choice. With two mandatory records, $B = 1$, $(\ell^A, \ell^H) = (5, 1)$ and $(0, 0.2)$, gain ordering reviews the first record (cost $1.05 + 1.5$) whereas the optimum reviews the second ($0.25 + 1.5$). The invariance to c_H also implies that the "fixed-route" hold-penalty sensitivity in the earlier draft was in fact an exact re-optimisation. Changing c_H shifts cost by $(|M| - B)^+ \Delta c_H / n$

without changing any route.

If optional-gain estimates satisfy $|\hat{g}_i - g_i| \leq \varepsilon$, the excess cost of the estimated optimum over the true optimum in regime (a) is at most $2(B-|M|)\varepsilon/n$. If reviewer-loss estimates satisfy $|\hat{\ell}_i^H - \ell_i^H| \leq \delta$, the excess in regime (b) is at most $2B\delta/n$. Both bounds follow by comparing each policy's estimated and true objective values. They make explicit that exact allocation does not guarantee good routing: routing quality is limited by estimation quality.

Proposition 2 (prioritisation does not change backlog). *Consider periods $t = 1, \ldots, T$ with new arrivals, capacity B per period, mandatory records carried forward while held, and the allocator of Proposition 1 applied each period to the union of backlog and arrivals. For any rule that orders mandatory records within a period, the number of holds, the backlog, the number of reviews and the number of completed records in every period are identical. Hence total waiting cost under a per-period holding charge is also identical.*

Proof. By induction on t. If the backlog L_t is the same across orderings, the mandatory count $|M_t| = L_t + |M_t^{new}|$ is the same, so holds $(|M_t| - B)^+$, mandatory reviews $\min(|M_t|, B)$ and slack for optional reviews are the same. L_{t+1} equals the number of holds and is therefore the same. Which optional records use the slack depends on $\hat{g}$, not on the mandatory ordering. □

Proposition 2 separates two managerial decisions. Staffing determines the size of the backlog. With mandatory arrival rate μ_t per record and capacity fraction b, a fluid approximation gives backlog growth of $(\mu_t - b)^+ n$ per period. Prioritisation determines only *which* records wait and the review-loss composition. By Little's law [41], the long-run mean waiting time then equals the average backlog divided by the review throughput. Section 5.9 tests both implications.

3.4. Two-head loss estimation from incomplete review logs

In practice reviewer feedback is sparse: only some records have been reviewed, and each logged review is a single realised decision rather than an expected loss. We therefore estimate (2) with two gradient-boosted regressors on a compact representation

$$z_i = (p_i,\ s_i,\ u_i,\ v_i,\ p_i u_i,\ p_i v_i,\ \hat{y}_i u_i,\ \hat{y}_i v_i), \qquad s_i = \min(c_{FN} p_i,\ c_{FP}(1 - p_i)), \tag{5}$$

where s_i is the Bayes risk of automation. The **automation head** $\hat{\ell}^A(z)$ is fitted on *all* labelled records of the gain-fitting partition with target ℓ_i^A. The **reviewer head** $\hat{\ell}^H(z)$ is fitted only on records with a logged review, with target $w(y_i)\,\mathbb{1}[\text{logged decision} \neq y_i]$. This target is an unbiased but noisy estimate of ℓ_i^H. The estimated gain is $\hat{g}_i = \hat{\ell}^A(z_i) - \hat{\ell}^H(z_i) - c_R$. Optional records are admitted only if $\hat{g}_i > 0$, and under overload mandatory records are ordered by $\hat{\ell}^H$ (Proposition 1b). The **single-head** alternative used in the earlier draft regresses $\ell_i^A - \ell_i^H - c_R$ directly and can therefore use only logged records. The representation has eight inputs regardless of dataset width. Probability-only (p_i, s_i), feature-aware (probability plus scaled raw features) and rich (probability, quality measures and raw values) versions are used as comparators.

Algorithm 1 summarises the routing procedure. Its cost is $O(n \log n)$ per batch, dominated by sorting. For each record the system stores a label-free audit tuple $(p_i, u_i, v_i, m_i, \hat{\ell}_i^A, \hat{\ell}_i^H, \text{action}_i)$. Fig. 1 shows the architecture.

Algorithm 1. Governance-constrained routing (per batch)

Input: records $\{x_i\}$; calibrated predictor; heads $\hat{\ell}^A, \hat{\ell}^H$; thresholds (τ_u, τ_v); capacity B.

1: for each record compute p_i, $\hat{y}_i$, u_i, v_i, z_i, m_i (Eqs. 1, 3, 5)

2: $\hat{g}_i \leftarrow \hat{\ell}^A(z_i) - \hat{\ell}^H(z_i) - c_R$

3: if $|M| \le B$: review all $i \in M$; among optional records with $\hat{g}_i > 0$ review the $B - |M|$ largest $\hat{g}_i$; automate the rest

4: else: review the B records in M with smallest $\hat{\ell}^H(z_i)$; hold the remaining $|M| - B$; automate all optional records

5: write audit tuple for every record; carry held records to the next batch

Output: actions and audit log.

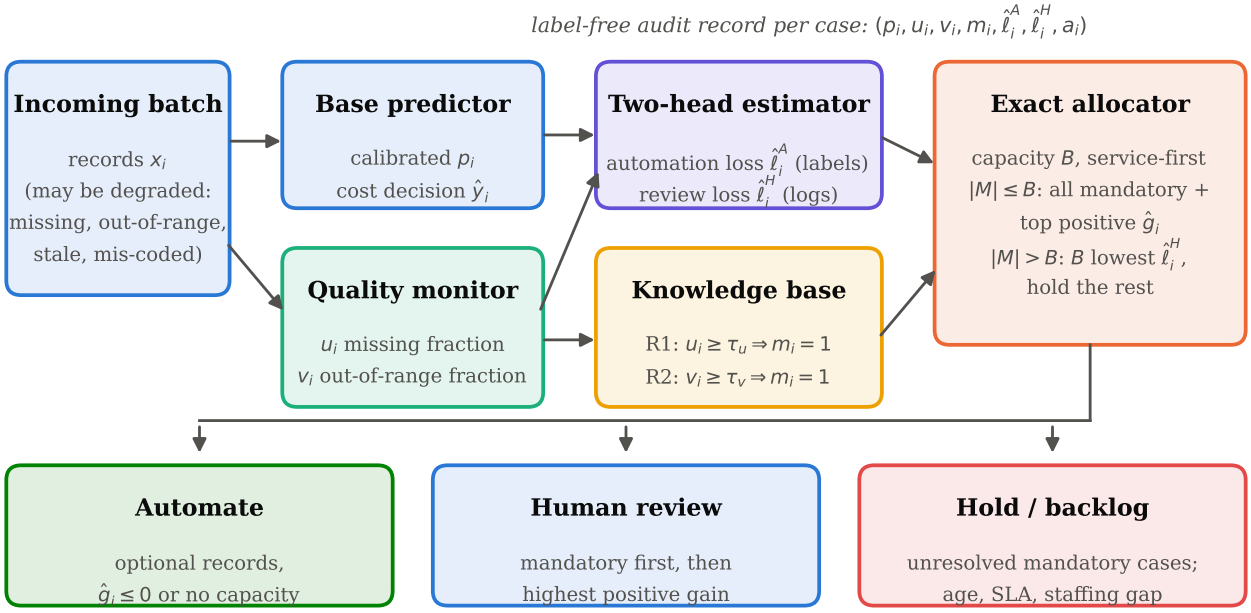


Fig. 1. Architecture of the governance-constrained routing system. The quality monitor feeds both the rule base and the loss estimator; the allocator enforces capacity and the service-first constraint, and held records form a backlog.

4. Experimental design

4.1. Datasets and partitions

Credit decisions are an established benchmarking domain [42]. We use three public credit datasets from the earlier draft and one external dataset: German numeric credit (1,000 records, 24 numeric predictors; class 2 recoded to 1) [43], Australian credit (690 records, 14 predictors) [44], default of credit-card clients (30,000 records, 23 predictors; identifier removed) [45], and HMEQ home-equity loans (5,960 records, 12 predictors, default indicator) [46]. HMEQ was not used during development. Unlike the other three datasets it contains naturally occurring missing values in 11 of 12 predictors, so it tests the method on real rather than injected incompleteness. Its two categorical fields are integer-coded.

Feature-identical records, including those with conflicting labels, are grouped before splitting. Groups are stratified by majority label into about 45% predictor training, 15% calibration, 20% gain fitting and 20% testing, so no feature vector crosses roles. Twenty seeds (2–21) define the evaluation splits. Seeds 0–1 were used during the original development and are excluded.

4.2. Base predictor and estimators

The base predictor is a histogram gradient-boosting classifier (at most 120 iterations, 15 leaves, learning rate 0.07, L2 penalty 1) trained on clean predictor-training rows. It handles missing values natively. Calibration is logistic regression on clipped log-odds, fitted on the calibration partition expanded into clean, missing, outlier and mixed versions. Loss heads are histogram gradient-boosting regressors (at most 100 iterations, seven leaves, learning rate 0.06, L2 penalty 5, minimum leaf size $\max(20, \lfloor N/200 \rfloor)$), fitted on the gain partition expanded in the same four ways. All bounds and medians come from predictor-training rows. Software: scikit-learn 1.9.1, NumPy 2.5 and SciPy 1.18.

4.3. Data-quality conditions

Test partitions are evaluated under nine conditions (Table 2). Family A reproduces the earlier protocol: defects that completeness and range checks can see. Family B adds domain-valid defects that are common in business data but largely invisible to such checks. All corruption is applied at test time, and Family B mechanisms never appear in calibration or gain fitting, so Family B also tests generalisation to unseen defect mechanisms. Code-like fields are those with at most ten distinct training values.

Table 2. Data-quality conditions. Family A defects are visible to completeness and range checks; Family B defects are domain-valid and largely invisible to them. Family B mechanisms are never used for fitting.

Family	Condition	Mechanism (per test record)	Visible to u, v?
—	clean	none (HMEQ: natural missingness)	—
A	missing	each cell missing with probability $r \sim U[0, 0.55]$	yes
A	outliers	cells with probability $r \sim U[0, 0.18]$ offset by $\pm U[1.2, 2.5]\, t_j$	mostly
A	mixed	outliers $r \sim U[0, 0.12]$, then missing $r \sim U[0, 0.35]$	yes
A	shift (severe)	outliers $r \sim U[0.08, 0.25]$, then missing $r \sim U[0.35, 0.75]$	yes
B	informative missing	missing rate $r \sim U[0, 0.35]$ scaled by 1.8 if $y = 1$, 0.6 if $y = 0$ (MNAR)	partly

Family	Condition	Mechanism (per test record)	Visible to u, v?
B	stale	with probability 0.5 a record's continuous fields are deflated by a common factor $U[0.55, 0.90]$ (outdated snapshot)	rarely
B	inconsistent codes	code-like cells with probability $r \sim U[0, 0.30]$ replaced by a different valid code	no
B	realistic mix	stale (0.3) + inconsistent ($U[0, 0.15]$) + informative missing ($U[0, 0.20]$)	partly

4.4. Reviewer models and feedback

Because no reviewer logs are available for these public datasets, reviewer behaviour is simulated. To avoid favouring the proposed representation we use five models (Table 3), only one of which depends on the router's quality signals. All are symmetric: the reviewer returns the wrong label with probability e_i.

Table 3. Simulated reviewer models. The feature-dependent and AI-correlated reviewers err on information the router does not observe; the AI-correlated reviewer errs most where the model errs.

Reviewer	Error probability e_i	Depends on router's (u, v)?
constant	0.10	no
quality-coupled (earlier draft)	$\min(0.40,\ 0.05 + 0.35u_i + 0.25v_i)$	yes
feature-dependent	$0.02 + 0.24\,\Phi(1.5\,\tilde{z}_i)$, where $\tilde{z}_i$ is a standardised hidden random projection of the *clean* features	no
AI-correlated	$\min(0.45,\ 0.05 + 0.25\,\mathbb{1}[\hat{y}_i \neq y_i] + 0.10(1 - \lvert 2p_i^{clean} - 1\rvert))$	no
adverse	0.25	no

The earlier draft trained gain models on expected reviewer losses for every record, which is optimistic. In the main protocol here (**logs**), a random 40% of gain-fitting records have a review log, and each log contains one sampled reviewer decision. Labels are available for all gain-fitting records. Every learned method is trained under this protocol. The optimistic protocol (**oracle targets**) is rerun for comparison, and the logged fraction is varied from 10% to 100% in Section 5.5.

4.5. Methods compared

All methods share the base predictor, the automated decision $\hat{y}_i$, the rule base, capacity and costs. They differ only in how they rank records for review (Table 4). Rule-constrained heuristics fill all remaining capacity. Learned methods review an optional record only if their own deferral criterion is positive. Because the published L2D methods have no notion of a hold, every method uses its own score to order mandatory records under overload,

and the overload-ordering rule is ablated separately. In the L2D baselines only the rejector is used; the automated decision remains the shared $\hat{y}_i$, so that differences reflect routing rather than classifier quality.

Table 4. Methods. L2D surrogates are one-hidden-layer networks (32 ReLU units, Adam, early stopping on a 20% validation split of logged records) implemented in NumPy, since a deep-learning framework was not required for tabular inputs.

Method	Ranking score	Information	Source / adaptation
Automation only	— (no review, rules ignored)	—	reference
Random + rules	uniform random	—	reference
Confidence	$-\lvert p_i - c_{FP}/(c_{FP}+c_{FN})\rvert$	p	boundary uncertainty
Chow cost risk	s_i	p	Chow's rule [7]; constant expert cost
Quality only	$u_i + v_i$	u, v	reference
Single-head gain (earlier draft)	$\hat{g}_i$ from one regressor	z_i	logged records only
L2D-CE	$g_\perp - \max_y g_y$ (softmax)	scaled x, u, v, p	Mozannar & Sontag [10], cost-sensitive weights $\max_j c_j - c_k$, MLP(32)
L2D-OvA	$\sigma(g_\perp) - \max_y \sigma(g_y)$	scaled x, u, v, p	Verma & Nalisnick [11], examples weighted by $w(y)$, MLP(32)
Differential triage	$\hat{\ell}^A - \hat{\ell}^H - c_R$, ordered by gain	p, u, v, raw x	Raghu et al. [13], two error models
Two-head, probability only	as proposed	p, s	quality ablation
Two-head, quality-aware (proposed)	$\hat{g}_i$; overload by $\hat{\ell}_i^H$	z_i	this paper
Oracle	true g_i, true ℓ_i^H	labels	upper bound, not deployable

4.6. Metrics, statistics and hypotheses

The primary metric is expected cost per record (Eq. 4), evaluated with true labels and expected reviewer losses. We also report review, hold and violation rates (a violation is an automated mandatory record). The primary comparison averages the three original datasets equally over the eight non-clean conditions and all five reviewer models at 20% capacity. Reviewer-specific and family-specific results are reported separately, and HMEQ is reported as an external check.

Repeated splits reuse observations, so ordinary paired t intervals are too narrow. We use the corrected resampled t-test of Nadeau and Bengio [47], which inflates the variance of the $J = 20$ split-level differences by $(1/J + n_{test}/n_{train})$ with $n_{test}/n_{train} = 0.25$. Holm's procedure [48] adjusts the family of planned comparisons. We also report the number of seeds (of 20) on which the proposed method has lower cost.

Before running the revised experiments we fixed four hypotheses:

- **H1 (coupled value).** Quality-aware two-head routing has lower cost than probability-only two-head routing when reviewer error depends on observable quality and defects are visible (quality-coupled reviewer, Family A).

- **H2 (decoupled null).** The advantage shrinks toward zero when reviewer error does not depend on observable quality or when defects are invisible (Family B).
- **H3 (overload ordering).** Under severe shift, ordering mandatory records by $\widehat{\ell}^H$ yields lower cost than ordering by $\widehat{g}$.
- **H4 (log efficiency).** With incomplete logs, the two-head estimator has lower cost than the single-head estimator, and the difference narrows as logs become complete.

4.7. Multi-period backlog simulation

To measure managerial outcomes we simulate 40 periods ("days") of 100 arrivals each, drawn with replacement from the test partition. Data quality is *mixed* on days 1–10 and 21–40 and *severe shift* on days 11–20, a ten-day deterioration episode. Held mandatory records are carried forward. The router is fitted once and not refitted. Capacity per day ranges from 10% to 70% of arrivals. We compare three prioritisation rules for mandatory records: first-in-first-out (FIFO), the cost-exact rule of Proposition 1b (lowest $\widehat{\ell}^H$ first), and the gain ordering of the earlier draft. Each held record incurs a waiting cost of 0.25 per day, and records unresolved at day 40 incur $c_H = 1.5$. Outcomes are cost per arrival, cost per completed case, completion rate, backlog, waiting time (mean and 95th percentile), the share of mandatory cases resolved within two days (SLA-2), reviewer override rate (sampled reviewer decision differs from the model decision), and capacity utilisation. Ten seeds per dataset are used, with the quality-coupled and feature-dependent reviewers.

5. Results

5.1. Replication of the earlier draft

We first reran the earlier protocol (oracle reviewer-loss targets, single-head gain models, quality-coupled reviewer, Family A, 20% capacity) in the reimplemented pipeline. Absolute values differ from the draft because the code was rebuilt from the manuscript description, but the pattern is the same. Quality-aware single-head gain cost 0.6121 versus 0.6150 for feature-aware gain, a paired difference of 0.0028 in its favour. The naive split interval was [−0.0009, 0.0066] and the corrected interval [−0.0064, 0.0120] (p = 0.53). The draft reported a difference of 0.0012 with interval [−0.0022, 0.0046]. As in the draft, the advantage came from the default dataset (0.6341 vs 0.6417) and reversed on Australian credit (0.5243 vs 0.5234). Confidence (0.6106) and Chow cost risk (0.6105) were again at least as good. Replacing oracle targets with noisy, incomplete logs raised the Family A cost of every learned method by 0.2–0.7% (0.6048→0.6085 for the two-head model) and left the heuristics unchanged, because they use no reviewer data.

5.2. Main comparison against published methods

Table 5 reports the primary comparison: three datasets, eight non-clean conditions and 20% capacity, under each reviewer model. All rule-constrained methods reviewed 19.8% of records, held 6.0% and had no rule violations, so differences in cost reflect *which* records were reviewed. Among methods that learn from review logs, the proposed two-head quality-aware router had the lowest macro cost (0.5355), ahead of the earlier single-head

design (0.5362), differential triage (0.5372), probability-only two-head (0.5381), L2D-OvA (0.5421) and L2D-CE (0.5461). The two simple cost-aware heuristics were lower still: Chow cost risk at 0.5291 and confidence at 0.5295. The oracle bound (0.3773) shows that most of the available improvement over these heuristics remains unrealised by every method.

Table 5. Expected cost per record at 20% capacity (lower is better), averaged over German, Australian and default datasets, eight non-clean conditions and 20 splits. Learned methods are trained on noisy, incomplete review logs. Bold marks the lowest deployable cost in each column. Automation only ignores the rules (12.6% of records violate them).

Method	Constant	Quality-coupled	Feature-dep.	AI-correlated	Adverse	Macro
Automation only (no rules)	0.5530	0.5530	0.5530	0.5530	0.5530	0.5530
Random + rules	0.5414	0.5638	0.5609	0.5689	0.6135	0.5697
Confidence	0.5033	0.5252	0.5191	**0.5375**	**0.5621**	0.5295
Chow cost risk	**0.5015**	**0.5242**	**0.5184**	0.5377	0.5640	**0.5291**
Quality only	0.5390	0.5787	0.5615	0.5695	0.6217	0.5741
Single-head gain (earlier draft)	0.5081	0.5282	0.5256	0.5454	0.5736	0.5362
L2D-CE [10]	0.5206	0.5435	0.5300	0.5527	0.5837	0.5461
L2D-OvA [11]	0.5152	0.5418	0.5274	0.5483	0.5777	0.5421
Differential triage [13]	0.5109	0.5326	0.5231	0.5455	0.5740	0.5372
Two-head, probability only	0.5108	0.5345	0.5265	0.5458	0.5729	0.5381
Two-head, quality-aware (proposed)	0.5070	0.5293	0.5250	0.5425	0.5736	0.5355
Oracle (uses labels; bound)	0.3470	0.3628	0.3539	0.4145	0.4084	0.3773

Table 6 gives the paired comparisons with corrected intervals. Against L2D-CE the proposed router was 1.95% cheaper on 19 of 20 splits (corrected p = 0.014), but this does not survive Holm adjustment across the eight planned comparisons (adjusted p = 0.11). No other difference was significant after correction. Against Chow cost risk and confidence the proposed method was about 1.2% *more* expensive and won on only five and four splits. We therefore do not claim superiority over simple cost-aware routing in the pooled comparison.

Table 6. Paired comparisons for the primary endpoint. Positive differences favour the proposed method. Intervals and p-values use the Nadeau–Bengio correction for overlapping training sets; Holm adjustment is over the seven comparisons shown plus automation only.

Proposed vs.	Δ cost (comparator – proposed)	Corrected 95% CI	Relative	Wins /20	p	Holm p
Single-head gain (earlier draft)	0.0007	[−0.0031, 0.0046]	+0.13%	13	0.70	1.00
L2D-CE	0.0106	[0.0024, 0.0189]	+1.95%	19	0.014	0.11

Proposed vs.	Δ cost (comparator − proposed)	Corrected 95% CI	Relative	Wins /20	p	Holm p
L2D-OvA	0.0066	[−0.0018, 0.0151]	+1.22%	17	0.12	0.70
Differential triage	0.0018	[−0.0064, 0.0100]	+0.33%	13	0.66	1.00
Two-head, probability only	0.0026	[−0.0025, 0.0078]	+0.49%	13	0.30	0.89
Chow cost risk	−0.0063	[−0.0144, 0.0017]	−1.20%	5	0.12	0.70
Confidence	−0.0060	[−0.0140, 0.0020]	−1.13%	4	0.13	0.70

Per dataset, the picture is consistent with a sample-size explanation (Table 7). On the default dataset, whose gain-fitting partition contains about 6,000 records, the proposed router beat every comparator on all 20 splits: Chow by 1.10%, confidence by 1.37%, L2D-CE by 1.29% and probability-only two-head by 1.45% (all corrected $p \leq 0.003$). On German (about 200 gain-fitting records) and Australian credit (about 140), it lost to Chow by 1.96% and 3.20%. Across all datasets and reviewer models, the learned methods' disadvantage relative to Chow grows with capacity (Fig. 2). This is expected: at higher capacity more marginal records must be ranked, and noisy gain estimates matter more there.

Table 7. Relative cost reduction of the proposed router by dataset at 20% capacity (positive favours the proposed method), pooled over eight non-clean conditions and five reviewers; wins out of 20 splits in parentheses. Bold: corrected $p < 0.01$.

Dataset (gain-fitting records)	vs. Chow cost risk	vs. Confidence	vs. L2D-CE	vs. Two-head prob. only
German (≈200)	−1.96% (3/20)	−0.98% (7/20)	+1.84% (16/20)	−0.07% (7/20)
Australian (≈140)	−3.20% (3/20)	−4.81% (1/20)	+2.94% (16/20)	+0.01% (9/20)
Default (≈6,000)	**+1.10% (20/20)**	**+1.37% (20/20)**	**+1.29% (20/20)**	**+1.45% (20/20)**
HMEQ, external (≈1,190)	+0.24% (11/20)	−2.22% (0/20)	+0.65% (14/20)	−0.03% (12/20)

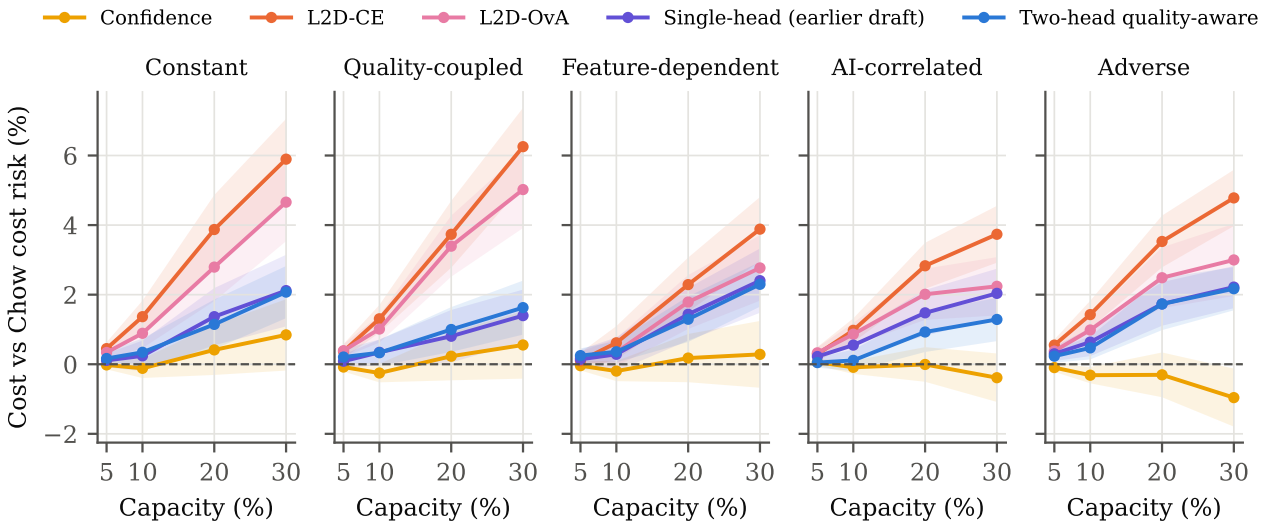


Fig. 2. Cost of learned routers relative to Chow cost risk across review capacity, by reviewer model (three datasets, eight non-clean conditions, 20 splits). Bands are ±1.96 standard errors across split means. Values above zero indicate higher cost than Chow's rule.

5.3. How much review data does a learned router need?

To test the sample-size explanation directly, we subsampled the gain-fitting partition of the default and HMEQ datasets while holding everything else fixed (Fig. 3). With about 200 gain-fitting records, the same size as the German partition, the proposed router was 1.96% more expensive than Chow on the default dataset and won on no split; L2D-CE was 5.06% more expensive. The gap closed at about 1,800 records (+0.10%, 10/20 wins) and reversed at 6,000 (−1.10%, 20/20 wins). The two-head quality-aware router was the best learned method at every size on the default dataset. Its advantage over the probability-only two-head router also grew with data (0.5 percentage points at 200 records, 1.5 at 6,000). HMEQ showed the same trend over its smaller range, from +2.47% at 119 records to −0.24% at 1,192. Because each gain-fitting record contributes four corrupted copies and 40% of copies carry a review log, 1,800 records correspond to roughly 2,900 logged review decisions. In this setting, then, a learned router needs a few thousand logged review outcomes before it beats a cost-aware heuristic. Below that volume Chow's rule is the better choice.

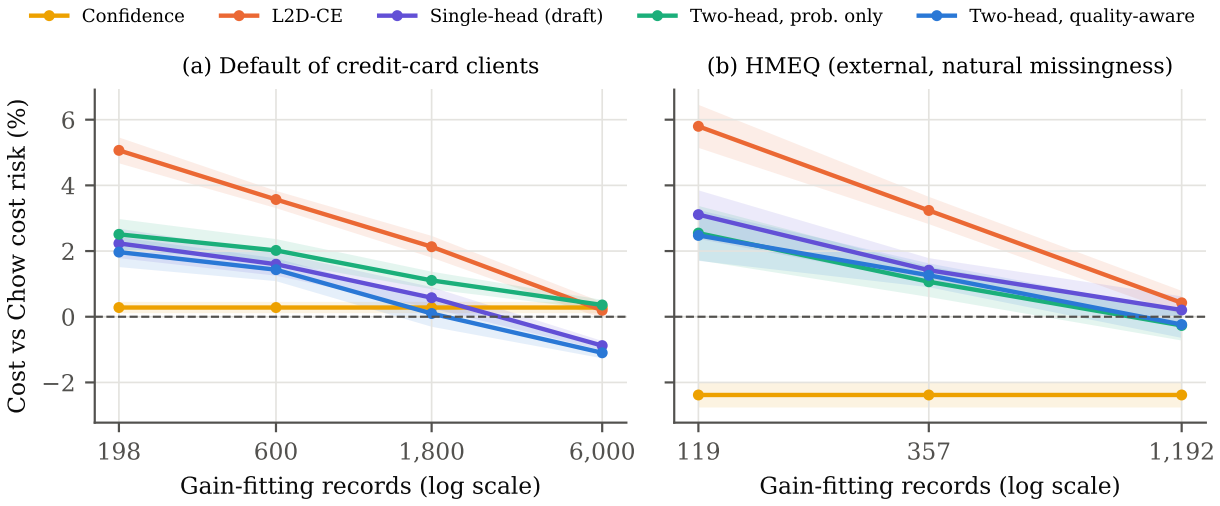


Fig. 3. Learned routing versus Chow's rule as the gain-fitting partition grows (20% capacity, eight non-clean conditions, five reviewers, 20 splits per size). Negative values indicate lower cost than Chow.

5.4. When does quality information help?

Fig. 4 reports the cost reduction from adding quality inputs (u, v and interactions) to the two-head router, holding everything else fixed, for every condition × reviewer cell at 20% capacity. The prespecified hypotheses fare as follows.

H1 (coupled value) was not supported at conventional significance. With the quality-coupled reviewer and visible defects, quality inputs reduced cost by 0.65% on 16 of 20 splits (corrected 95% CI [−0.0029, 0.0108], p = 0.24).

H2 (decoupled null) was supported for visible defects but not for invisible ones. When reviewer error did not depend on quality and defects were visible, the effect was essentially zero: −0.07%, 10/20 wins, p = 0.80. Contrary to H2, quality inputs helped *more* under Family B (domain-valid defects): +1.41% with the coupled reviewer and +0.94% with the decoupled reviewers, neither significant. The effect came almost entirely from one condition. Under informative missingness, quality inputs reduced cost under every reviewer model, from +1.2% (adverse) to +3.5% (constant; 17/20, p = 0.039). Pooled over reviewers the reduction was 2.29% (18/20, p = 0.045).

The mechanism is identified by routing quality information to only one head (Table 8). Under informative missingness, quality inputs in the automation-loss head alone reduced cost by 2.83% on all 20 splits (p = 0.013), whereas quality inputs in the reviewer-loss head alone did nothing (−0.28%, 7/20). Missingness that depends on the outcome makes the base predictor's error on incomplete records predictable from the missing fraction. The quality signal therefore improves the estimate of *how wrong the automated decision is likely to be*, not of how accurate the reviewer will be. Through the reviewer-loss head, quality inputs helped only when reviewer error actually depended on quality: +0.70% (16/20, p = 0.17) with the coupled reviewer on Family A, versus −0.41% (7/20) with the feature-dependent reviewer. The earlier draft's concern that its reviewer simulation favoured the

quality representation was therefore justified in direction but small in magnitude, and this channel is not where most of the value lies.

Table 8. Where quality information acts. Cost reduction relative to the probability-only two-head router at 20% capacity (positive = quality inputs help); wins out of 20 splits. Bold: corrected p < 0.05.

Condition	Quality in automation head only	Quality in reviewer head only	Quality in both (proposed)
Informative missing (all reviewers)	**+2.83% (20/20)**	−0.28% (7/20)	**+2.29% (18/20)**
Stale records (all reviewers)	+0.98% (13/20)	+0.04% (10/20)	+0.91% (14/20)
Realistic mix (all reviewers)	+1.86% (15/20)	−0.74% (6/20)	+0.91% (14/20)
Family A, quality-coupled reviewer	+0.07% (13/20)	+0.70% (16/20)	+0.65% (16/20)
Family A, feature-dependent reviewer	+0.18% (11/20)	−0.41% (7/20)	−0.34% (4/20)

Stale records and inconsistent codes, which the quality monitor cannot see (correlation between the true defect fraction and $u + v$ of −0.01 to 0.14 for stale records and 0.14 to 0.45 for inconsistent codes), produced small and unstable effects. They also produced almost no mandatory reviews: at most 0.5% of records were flagged, against 13–18% under the missing condition.

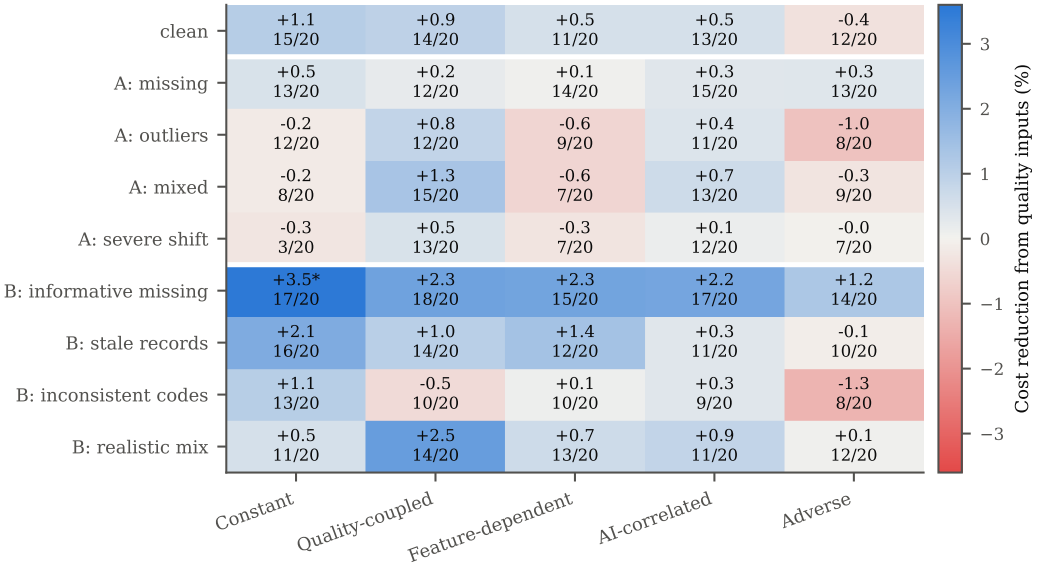


Fig. 4. Cost reduction (%) from adding quality inputs to the two-head router, by data condition and reviewer model, at 20% capacity. Each cell shows the mean over three datasets and the number of splits (of 20) on which quality inputs helped; * marks corrected p < 0.05. Blue: quality helps; red: quality hurts.

5.5. Ablations: rules, overload ordering and log completeness

Rules versus quality. Table 9 separates the quality inputs from the rule layer. In the mixed and realistic conditions the rule layer added 0.7–1.7% to cost (for example 0.4614→0.4679 under mixed), the price of reviewing flagged

records instead of higher-gain optional ones. Removing the rules left 2.1% and 0.7% of records automated in violation. Under severe shift the rules forced 47.5% of records into holds and raised cost by 67% (0.5987→1.0015). Without rules the router automated 54% of records the policy deems unfit for automation. The two factors were roughly additive: the quality effect had the same sign with and without rules in all three conditions.

Table 9. 2 × 2 ablation of quality inputs and the rule layer for the two-head router (three datasets, five reviewers, 20 splits, 20% capacity). Unconstrained rows are diagnostic: they are cheaper because they automate records the governance policy forbids automating.

Condition	Quality inputs	Rules	Cost	Review %	Hold %	Violation %
Mixed	off	on	0.4688	19.8	0.0	0.0
Mixed	on	on	0.4679	19.8	0.0	0.0
Mixed	off	off	0.4610	19.8	0.0	2.2
Mixed	on	off	0.4614	19.8	0.0	2.1
Realistic mix	off	on	0.4782	19.8	0.0	0.0
Realistic mix	on	on	0.4739	19.8	0.0	0.0
Realistic mix	off	off	0.4748	19.8	0.0	0.7
Realistic mix	on	off	0.4687	19.8	0.0	0.7
Severe shift	off	on	1.0016	19.8	47.5	0.0
Severe shift	on	on	1.0015	19.8	47.5	0.0
Severe shift	off	off	0.6028	19.7	0.0	54.5
Severe shift	on	off	0.5987	19.6	0.0	54.2

Overload ordering (H3) was not supported empirically. Ordering mandatory records by $\widehat{\ell}^H$ (Proposition 1b) rather than by $\hat{g}$ reduced cost under severe shift by only 0.18% (14/20, p = 0.47); with the quality-coupled reviewer the reduction was 0.47% (p = 0.30). The correction is exact for true losses, but under severe shift almost all mandatory records have similar estimated reviewer loss, so the two orderings select largely the same records. Section 5.9 shows that in a multi-period system the ordering affects waiting times far more than cost.

Log completeness (H4) received only directional support. In the primary comparison the two-head estimator was 0.13% cheaper than the single-head estimator (13/20, p = 0.70). In a dedicated sweep (three datasets, mixed and realistic conditions, quality-coupled and feature-dependent reviewers, 10 splits), the two-head estimator was 0.60% cheaper with 10% of records logged and 0.54% cheaper with 40%, and 0.46% more expensive with complete logs (all p > 0.5). This is the direction H4 predicts, since the two-head design's advantage is that its automation head can use unlogged records, but the effect is too small to establish with these data.

5.6. External dataset with natural missingness

On HMEQ, where 43.6% of records have at least one naturally missing value and DEBTINC is missing for 21.3% of records, the quality-aware router was evaluated without any injected corruption (Table 10, "natural" column). There it was cheaper than the probability-only two-head router (0.2053 vs 0.2163; 19/20 splits, corrected p = 0.019) and than Chow cost risk (0.2288; 19/20, p < 0.001). This matches the informative-missingness mechanism above: natural missingness in HMEQ is associated with default. Confidence ranking was cheaper still (0.1847; the proposed router won on 2 of 20 splits). Under the rules, 3.3% of HMEQ records were mandatory in the natural

condition. Pooled over all nine HMEQ conditions, confidence (0.3950) and differential triage (0.4027) had the lowest deployable costs, and the proposed router (0.4056) was 0.9% below Chow (0.4091).

Table 10. HMEQ external dataset at 20% capacity (expected cost per record, five reviewers, 20 splits). The outliers column is omitted for space but included in the mean. "Natural" applies no injected corruption.

Method	Natural	Missing	Mixed	Shift	Informative	Stale	Inconsistent	Realistic	Mean (9)
Confidence	**0.1847**	0.5148	0.4135	1.0844	0.3319	**0.2223**	**0.2038**	0.2957	**0.3950**
Chow cost risk	0.2288	0.5247	0.4137	1.0871	0.3333	0.2506	0.2363	0.3029	0.4091
Single-head gain (earlier draft)	0.2065	0.5228	0.4192	1.0905	0.3444	0.2389	0.2226	0.3153	0.4074
L2D-CE	0.2182	0.5212	0.4215	1.0911	0.3343	0.2406	0.2336	0.3065	0.4095
L2D-OvA	0.2118	0.5171	0.4210	1.0890	0.3338	0.2367	0.2320	0.3048	0.4071
Differential triage	0.2082	0.5199	**0.4107**	1.0837	0.3326	0.2434	0.2227	0.3011	0.4027
Two-head, probability only	0.2163	0.5178	0.4188	**1.0826**	0.3407	0.2447	0.2287	0.3049	0.4067
Two-head, quality-aware	0.2053	0.5187	0.4176	1.0868	0.3450	0.2370	0.2208	0.3153	0.4056

5.7. Governance-rule alternatives

Table 11 compares four rule sets for the proposed router, with the router unchanged. Because the rules are evaluated against their own thresholds, each rule set has zero violations by construction. To judge whether a rule set is *appropriate*, we report the share of truly defective records (more than 20% of cells corrupted) that were nevertheless automated. Under severe shift, the main rules reduced automated defective records from 80.3% (no rules) to 32.2% at a cost increase of 71%, and strict rules reduced them to 4.5% at a cost increase of 111%, with 75.6% of records held. Under the realistic mix, whose defects are mostly invisible, even strict rules left 74.7% of defective records automated, because rules built on completeness and range checks cannot see stale or mis-keyed values. Lenient rules were nearly indistinguishable from no rules except under shift. Rule sets shifted all ranking methods by similar amounts. Confidence had the lowest cost under the none, lenient and main rule sets, with Chow second. Under strict rules, where three-quarters of records are mandatory, confidence, L2D-OvA, the proposed router and its rich-input variant were within 0.001 of each other and all slightly below Chow.

Table 11. Governance-rule alternatives for the proposed router (German, Australian and default; quality-coupled and feature-dependent reviewers; 10 splits; 20% capacity). "Defective records automated" is the share of records with more than 20% corrupted cells that the policy automated. The lenient row for the realistic mix equals "none" and is omitted.

Condition	Rule set (τ_u, τ_v)	Mandatory %	Cost	Hold %	Defective records automated %
Mixed	none	0.0	0.4415	0.0	80.9
Mixed	lenient (0.7, 0.4)	0.1	0.4424	0.0	80.9
Mixed	main (0.5, 0.25)	2.8	0.4533	0.0	79.3
Mixed	strict (0.3, 0.15)	23.4	0.5268	3.9	60.0
Severe shift	none	0.0	0.5888	0.0	80.3
Severe shift	lenient	19.1	0.6437	1.7	78.5
Severe shift	main	67.8	1.0093	48.0	32.2
Severe shift	strict	95.5	1.2413	75.6	4.5
Realistic mix	none	0.0	0.4395	0.0	80.8
Realistic mix	main	0.8	0.4449	0.0	80.3
Realistic mix	strict	6.4	0.4763	0.0	74.7

5.8. Cost settings with re-optimisation

In contrast to the earlier draft's fixed-route sensitivity, here every cost change refits the decision threshold, the loss heads and the L2D surrogates (Table 12). The ranking was stable. Confidence had the lowest cost in all seven settings, Chow was second, and the proposed router was third, 0.4–1.2% above Chow. It remained ahead of the probability-only router, the single-head router and both L2D surrogates in every setting. Changing the hold penalty from 0.5 to 3.0 raised the proposed router's cost by 0.3998, equal to 2.5 × its 16.0% hold rate, with identical review and hold rates. This confirms empirically the invariance in Proposition 1: under the service-first constraint, the hold penalty changes the price of overload but not the routes.

Table 12. Expected cost after re-optimising all methods for each cost setting (German, Australian and default; mixed, shift and realistic conditions; two reviewers; 10 splits; 20% capacity). Small datasets dominate this average; see Table 7 for the dataset-size dependence.

Setting	Chow	Confidence	Proposed	Two-head prob.	Single-head	L2D-CE	L2D-OvA
FN:FP = 2:1	0.4877	**0.4874**	0.4925	0.4946	0.4945	0.5025	0.5008
FN:FP = 5:1 (main)	0.6320	**0.6314**	0.6358	0.6406	0.6384	0.6476	0.6444
FN:FP = 10:1	0.7216	**0.7157**	0.7302	0.7323	0.7339	0.7446	0.7359
Review fee 0	0.6221	**0.6215**	0.6259	0.6307	0.6285	0.6380	0.6345
Review fee 0.20	0.6618	**0.6612**	0.6655	0.6703	0.6681	0.6788	0.6741
Hold penalty 0.5	0.4721	**0.4715**	0.4759	0.4806	0.4785	0.4877	0.4844
Hold penalty 3.0	0.8720	**0.8714**	0.8757	0.8805	0.8783	0.8875	0.8843

5.9. Managerial outcomes in a multi-period system

Table 13 and Fig. 5 report the 40-day simulation (four datasets, two reviewers, ten seeds). Mandatory demand averaged 5.5% of arrivals in mixed periods and 70.0% during the ten-day severe-shift episode. As Proposition 2 predicts, the three prioritisation rules produced identical backlogs, completion rates and capacity utilisation at every staffing level, and cost per arrival differed by at most 0.005. The fluid approximation predicted the peak backlog closely: 500, 300 and 200 records at 20%, 40% and 50% capacity, against simulated means of 499.8, 299.7 and 199.7 (Fig. 5a). At 20% capacity, 5.2% of all arrivals were still unresolved at day 40, and cost per arrival was 1.16 against 0.39 at 70% capacity. Capacity, not routing, dominated cost during the deterioration episode.

Prioritisation changed *who* waited. FIFO ordering cut the 95th-percentile wait at 40% capacity from 13.8 days (cost-exact ordering) to 7.1 days and the maximum wait from 18.4 to 8.0 days. But it reduced the share of mandatory cases resolved within two days from 65.3% to 41.1%, because FIFO serves old cases first and newly flagged cases wait. The cost-exact and gain orderings behaved almost identically, again because estimated reviewer losses vary little among mandatory records. Reviewers overturned the automated decision in 44–49% of reviewed cases. With a 5:1 cost ratio, the automated decision deliberately labels many uncertain records as positive, so a high override rate is expected and is not by itself evidence of a poor model.

Table 13. Multi-period managerial outcomes (40 days, 100 arrivals per day, severe-shift episode on days 11–20; four datasets, two reviewers, ten seeds). The gain ordering of the earlier draft is omitted; it differs from the cost-exact rule by at most 0.5 days in any waiting-time measure and at most 0.04 in any rate.

Capacity	Order	Cost / arrival	Cost / completed	Completed %	Peak backlog	Mean wait (d)	P95 wait (d)	SLA-2 %	Override %
20%	FIFO	1.161	1.148	94.8	499.8	9.8	20.2	18.5	48.3
20%	Cost-exact	1.157	1.144	94.8	499.8	7.5	23.0	48.4	46.8
40%	FIFO	0.634	0.634	100.0	299.7	3.4	7.1	41.1	48.2
40%	Cost-exact	0.634	0.634	100.0	299.7	3.4	13.8	65.3	48.2
50%	FIFO	0.516	0.516	100.0	199.7	1.7	3.9	71.8	47.3
50%	Cost-exact	0.516	0.516	100.0	199.7	1.7	10.0	78.4	47.3
70%	FIFO	0.389	0.389	100.0	26.8	0.1	0.5	100.0	45.6
70%	Cost-exact	0.389	0.389	100.0	26.8	0.1	0.9	98.2	45.6

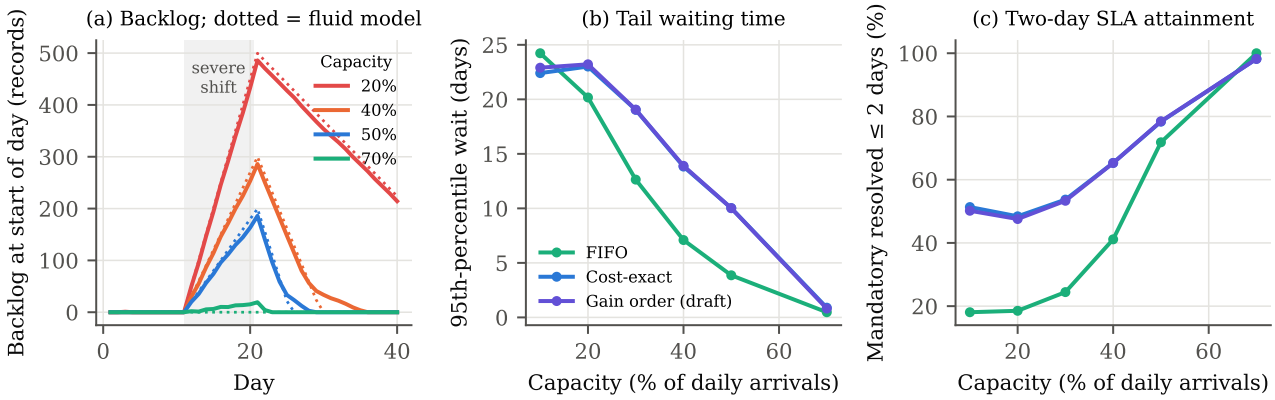


Fig. 5. Multi-period simulation. (a) Backlog over time (seed 2, quality-coupled reviewer, mean of four datasets) at four staffing levels, with the fluid approximation dotted; the shaded band is the severe-shift episode. (b) 95th-percentile waiting time and (c) share of mandatory cases resolved within two days, by prioritisation rule, averaged over four datasets, two reviewers and ten seeds.

6. Discussion

6.1. Answer to the research question

The research question asked under what data-quality and review-capacity conditions quality-aware deferral improves decision cost and service completion while satisfying explicit governance rules. The evidence supports a conditional answer with four parts.

First, **quality information pays when missingness is informative about the outcome.** In that case it improves the estimate of automated-decision error, not of reviewer error. This was the only condition with a consistent effect across all five reviewer models (+1.2% to +3.5%), and it reappeared on the external HMEQ data, where missingness is natural rather than injected. When defects were visible but uninformative about the outcome, or invisible to completeness and range checks, quality inputs added little and sometimes hurt.

Second, **learned routing pays only with enough review history.** Holding information, rules and costs fixed, every learned router, including the published L2D surrogates, was worse than Chow's cost-risk rule when trained on a few hundred records. The proposed router overtook Chow at roughly 1,800 gain-fitting records (about 2,900 logged review decisions) and beat every comparator on all splits at 6,000. The pooled comparison in Table 5, in which two of three datasets are small, therefore understates what a learned router achieves at operational data volumes and overstates it for small portfolios.

Third, **capacity, not ranking, decides service completion under deterioration.** When mandatory demand exceeded capacity, the backlog was fully determined by staffing (Proposition 2), and the fluid rule $(\mu - b)^{+}n$ predicted the peak backlog to within 0.3 records. Ranking methods moved cost by about 1%; moving capacity from 20% to 40% during the episode cut cost per arrival by 45% and eliminated unresolved cases.

Fourth, **governance rules built on observable quality checks protect against visible defects only.** They prevented the automation of most severely degraded records under visible shift, at a substantial service cost, but let

three-quarters of records with invisible defects through even under strict thresholds.

6.2. Implications for knowledge-based decision systems

The study separates three components that are often conflated in deferral systems: a *knowledge layer* that encodes non-negotiable policy, an *estimation layer* that predicts the value of review, and an *allocation layer* that enforces capacity. Proposition 1 shows that once the policy is fixed and estimates are given, allocation is a sorting problem with an exact solution in both regimes. The design effort should therefore go into estimation and into the rules themselves, not into the allocator. The mechanism analysis (Table 8) adds a design principle: quality signals belong in the model of *automated* error when missingness may be informative, and in the model of *reviewer* error only when there is evidence that reviewers are affected by the same defects. The two-head decomposition makes that choice explicit and auditable, and lets the automation head use all labelled outcomes even when review logs are sparse.

The results also qualify the value of methodological sophistication in this setting. Two consistent L2D surrogates, adapted only by adding cost weights and a shared classifier, were the weakest learned methods at small and medium data volumes. This is unsurprising given that they learn a three-way decision from scratch, whereas the two-head router and Chow's rule start from a calibrated probability, but it suggests that reviews of deferral methods should report sample-size dependence alongside asymptotic guarantees.

6.3. Implications for managers

Four operational recommendations follow, each tied to a measured outcome:

1. **Start with a cost-aware heuristic.** Until a few thousand review outcomes have been logged, route by Bayes risk of the automated decision (Chow's rule) or by closeness to the cost threshold. Both use no reviewer data, cannot overfit review logs, and were within about 1% of the best learned router in most settings.
2. **Staff to mandatory demand, not to average demand.** Review capacity must exceed the mandatory arrival rate during deterioration episodes, or a backlog of $(\mu - b)^+$ per record per period accumulates. In our simulation a ten-day episode at 70% mandatory demand left about 500 records waiting at 20% capacity and about 27 at 70%. Monitoring the mandatory rate is a leading indicator of staffing gaps.
3. **Choose the queue discipline for its service profile, not its cost.** FIFO halves tail waiting time; cost-ordered queues resolve more cases within two days but let some cases wait more than two weeks. The choice belongs in the service-level agreement.
4. **Do not treat override rates or zero violations as evidence of good governance.** With asymmetric costs, reviewers overturned nearly half of automated decisions by design. Zero rule violations held by construction even when three-quarters of defective records were automated under invisible defects. Governance reporting should include defect-detection coverage, unresolved-case age and the mandatory-to-capacity ratio.

7. Limitations and future work

Simulated reviewers. No public dataset links credit decisions to logged human reviews, so reviewer behaviour is simulated. We reduced dependence on any single assumption by using five reviewer models, two of which err on information the router cannot see, and by training on sampled rather than expected reviewer outcomes. Real reviewers may also learn, tire, anchor on the model's recommendation [30] or change behaviour under backlog pressure, none of which is modelled. Validation with operational review logs is the most important next step.

Datasets and corruption. The three original benchmarks are old and small, and Family B mechanisms are stylised versions of real defects. HMEQ provides natural missingness but is itself a public benchmark of uncertain provenance. Category codes are treated as integers, and no contemporary dataset could be obtained in the computing environment used for this revision. Results for invisible defects depend on how stale and mis-keyed values are generated.

Rules and costs. Thresholds, the 5:1 cost ratio, the review fee and the hold penalty are illustrative rather than elicited from an organisation. We varied each and refitted the policies, and the method ranking was stable, but appropriate values must come from domain owners and regulators. The defect-based appropriateness measure in Table 11 uses an arbitrary 20% threshold.

Statistics. Repeated splits reuse observations. The corrected resampled t-test reduces but does not eliminate the resulting optimism, and with three primary datasets, dataset-level generalisation rests largely on the data-volume and external analyses. Several effects of interest (H1, H3, H4) were directionally consistent but not significant, and should be treated as inconclusive rather than as evidence of no effect.

Scope of the method. The allocator is exact for the stated single-batch problem, but the multi-period system uses it myopically, without anticipating future arrivals. Congestion-aware deferral [28] and constraint-programming assignment [15] address related dynamic and multi-expert problems. The L2D baselines were adapted (shared classifier, cost weights, a small network) and may perform better with tuning or larger architectures. Fairness, legal sufficiency of the rules and causal effects of review on outcomes are outside the scope of this study.

8. Conclusion

This study re-examined a data-quality-aware deferral system with explicit governance rules and review capacity, against published deferral methods, under reviewer models that do not favour its representation and under defects that quality checks cannot see. The system's allocator solves the governance-constrained routing problem exactly in both feasible and overloaded regimes, and its two-head estimator learns from incomplete review logs. It did not, however, outperform simple cost-aware routing in the pooled comparison dominated by small datasets. Its value is conditional. It outperformed all comparators when trained on several thousand records, and quality information improved it chiefly when missingness was informative about outcomes, through better estimates of automated error rather than of reviewer error. Under severe data deterioration, staffing determined backlog and service completion far more than any routing choice, and quality-triggered rules protected only against the defects they could observe. For knowledge-based decision systems used in AI governance, the practical message is to

make the rule layer, the estimation layer and the capacity layer separately visible, to measure each, and to reserve learned routing for settings with sufficient review history.

Data and code availability

All data are public: German, Australian and default credit from the UCI Machine Learning Repository [43–45] and HMEQ [46]. The code, configuration, seed-level results for every experiment, and scripts that regenerate every table and figure are in the `budgeted_deferral/` directory of the author's repository https://github.com/ykunam/data-quality-budgeted-deferral. A versioned release with an archival DOI should be created before submission.

Declaration of competing interest

The author must confirm before submission. No funding or competing interests are asserted in this draft.

Funding

To be confirmed by the author before submission.

CRediT authorship contribution statement

Yeshwanth Reddy Kunam: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization. (To be confirmed by the author.)

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

ChatGPT (OpenAI) assisted with the original study design, code and drafting. Claude (Anthropic) assisted with the revision: reimplementing the experimental pipeline, implementing the comparison methods, running the experiments, statistical analysis, figures, literature checking and manuscript drafting. The author must critically review, verify and adopt all content before submission and takes full responsibility for the accuracy and originality of the published work.

References

[1] R. Abraham, J. Schneider, J. vom Brocke, Data governance: A conceptual framework, structured review, and research agenda, Int. J. Inf. Manag. 49 (2019) 424–438. https://doi.org/10.1016/j.ijinfomgt.2019.07.008

[2] M. Janssen, P. Brous, E. Estevez, L.S. Barbosa, T. Janowski, Data governance: Organizing data for trustworthy artificial intelligence, Gov. Inf. Q. 37 (2020) 101493. https://doi.org/10.1016/j.giq.2020.101493

[3] E. Papagiannidis, P. Mikalef, K. Conboy, Responsible artificial intelligence governance: A review and research framework, J. Strateg. Inf. Syst. 34 (2025) 101885. https://doi.org/10.1016/j.jsis.2024.101885

[4] B.W. Wirtz, J.C. Weyerer, I. Kehl, Governance of artificial intelligence: A risk and guideline-based integrative framework, Gov. Inf. Q. 39 (2022) 101685. https://doi.org/10.1016/j.giq.2022.101685

[5] European Parliament and Council, Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), Off. J. Eur. Union L, 2024/1689, 12 July 2024, Art. 14 and Annex III(5)(b).

[6] B. Green, The flaws of policies requiring human oversight of government algorithms, Comput. Law Secur. Rev. 45 (2022) 105681. https://doi.org/10.1016/j.clsr.2022.105681

[7] C. Chow, On optimum recognition error and reject tradeoff, IEEE Trans. Inf. Theory 16 (1970) 41–46. https://doi.org/10.1109/TIT.1970.1054406

[8] X.Y. Zhang, G.S. Xie, X. Li, T. Mei, C.L. Liu, A survey on learning to reject, Proc. IEEE 111 (2023) 185–215. https://doi.org/10.1109/JPROC.2023.3238024

[9] D. Madras, T. Pitassi, R. Zemel, Predict responsibly: Improving fairness and accuracy by learning to defer, in: Adv. Neural Inf. Process. Syst. 31 (2018).

[10] H. Mozannar, D. Sontag, Consistent estimators for learning to defer to an expert, in: Proc. 37th Int. Conf. Mach. Learn., PMLR 119 (2020) 7076–7087.

[11] R. Verma, E. Nalisnick, Calibrated learning to defer with one-vs-all classifiers, in: Proc. 39th Int. Conf. Mach. Learn., PMLR 162 (2022) 22184–22202.

[12] H. Mozannar, H. Lang, D. Wei, P. Sattigeri, S. Das, D. Sontag, Who should predict? Exact algorithms for learning to defer to humans, in: Proc. 26th Int. Conf. Artif. Intell. Stat., PMLR 206 (2023) 10520–10545.

[13] M. Raghu, K. Blumer, G. Corrado, J. Kleinberg, Z. Obermeyer, S. Mullainathan, The algorithmic automation problem: Prediction, triage, and human effort, arXiv preprint (2019). https://doi.org/10.48550/arXiv.1903.12220

[14] N. Okati, A. De, M. Gomez-Rodriguez, Differentiable learning under triage, in: Adv. Neural Inf. Process. Syst. 34 (2021). https://doi.org/10.48550/arXiv.2103.08902

[15] J.V. Alves, D. Leitão, S. Jesus, M.O.P. Sampaio, J. Liébana, P. Saleiro, M.A.T. Figueiredo, P. Bizarro, Cost-sensitive learning to defer to multiple experts with workload constraints, Trans. Mach. Learn. Res. (2024). https://openreview.net/forum?id=TAvGZm2Rqb

[16] P. Hemmer, L. Thede, M. Vössing, J. Jakubik, N. Kühl, Learning to defer with limited expert predictions, Proc. AAAI Conf. Artif. Intell. 37 (2023) 6002–6011. https://doi.org/10.1609/aaai.v37i5.25742

[17] R.Y. Wang, D.M. Strong, Beyond accuracy: What data quality means to data consumers, J. Manag. Inf. Syst. 12 (1996) 5–33. https://doi.org/10.1080/07421222.1996.11518099

[18] E. Breck, N. Polyzotis, S. Roy, S.E. Whang, M. Zinkevich, Data validation for machine learning, in: Proc. Mach. Learn. Syst. 1 (2019) 334–347.

[19] C. Chai, J. Wang, Y. Luo, Z. Niu, G. Li, Data management for machine learning: A survey, IEEE Trans. Knowl. Data Eng. 35 (2023) 4646–4667. https://doi.org/10.1109/TKDE.2022.3148237

[20] Board of Governors of the Federal Reserve System, Office of the Comptroller of the Currency, Supervisory guidance on model risk management, SR Letter 11-7, 4 April 2011.

[21] Y. Geifman, R. El-Yaniv, SelectiveNet: A deep neural network with an integrated reject option, in: Proc. 36th Int. Conf. Mach. Learn., PMLR 97 (2019) 2151–2159.

[22] J. Mena, O. Pujol, J. Vitrià, Uncertainty-based rejection wrappers for black-box classifiers, IEEE Access 8 (2020) 101721–101746. https://doi.org/10.1109/ACCESS.2020.2996495

[23] R. Verma, D. Barrejón, E. Nalisnick, Learning to defer to multiple experts: Consistent surrogate losses, confidence calibration, and conformal ensembles, in: Proc. 26th Int. Conf. Artif. Intell. Stat., PMLR 206 (2023) 11415–11434.

[24] A. Mao, M. Mohri, Y. Zhong, Mastering multiple-expert routing: Realizable H-consistency and strong guarantees for learning to defer, in: Proc. 42nd Int. Conf. Mach. Learn., PMLR 267 (2025) 43035–43066.

[25] S. Liu, Y. Cao, Q. Zhang, L. Feng, B. An, Mitigating underfitting in learning to defer with consistent losses, in: Proc. 27th Int. Conf. Artif. Intell. Stat., PMLR 238 (2024) 4816–4824.

[26] Z. Wei, Y. Cao, L. Feng, Exploiting human-AI dependence for learning to defer, in: Proc. 41st Int. Conf. Mach. Learn., PMLR 235 (2024) 52484–52499.

[27] G. DeSalvo, C. Mohri, M. Mohri, Y. Zhong, Budgeted multiple-expert deferral, arXiv preprint (2025). https://doi.org/10.48550/arXiv.2510.26706

[28] V. Saketos, W. Reveillard, A. Proutiere, S. Dalleiger, M. Xiao, To defer or not to defer? Online learning with LLMs and congested experts, arXiv preprint (2026). https://doi.org/10.48550/arXiv.2610.15128

[29] S. Tariq, M. Baruwal Chhetri, S. Nepal, C. Paris, A2C: A modular multi-stage collaborative decision framework for human–AI teams, Expert Syst. Appl. 282 (2025) 127318. https://doi.org/10.1016/j.eswa.2025.127318

[30] G. Bansal, B. Nushi, E. Kamar, E. Horvitz, D.S. Weld, Is the most accurate AI the best teammate? Optimizing AI for teamwork, Proc. AAAI Conf. Artif. Intell. 35 (2021) 11405–11414. https://doi.org/10.1609/aaai.v35i13.17359

[31] A. Farinhas, N.M. Guerreiro, S. Agrawal, R. Rei, A. Martins, Translate smart, not hard: Cascaded translation systems with quality-aware deferral, in: Proc. 2025 Conf. Empirical Methods Nat. Lang. Process. (2025) 26742–26756. https://doi.org/10.18653/v1/2025.emnlp-main.1358

[32] L. Zhang, D. Jeong, S. Lee, Data quality management in the Internet of Things, Sensors 21 (2021) 5834. https://doi.org/10.3390/s21175834

[33] R.J.A. Little, D.B. Rubin, Statistical Analysis with Missing Data, 3rd ed., Wiley, Hoboken, NJ, 2019. https://doi.org/10.1002/9781119482260

[34] G. Sklavos, G. Theodossiou, Z. Papanikolaou, C. Karelakis, K. Ragazou, Environmental, social, and governance-based artificial intelligence governance: Digitalizing firms' leadership and human resources management, Sustainability 16 (2024) 7154. https://doi.org/10.3390/su16167154

[35] M.H. Jarrahi, Artificial intelligence and the future of work: Human-AI symbiosis in organizational decision making, Bus. Horiz. 61 (2018) 577–586. https://doi.org/10.1016/j.bushor.2018.03.007

[36] P. Mikalef, M. Gupta, Artificial intelligence capability: Conceptualization, measurement calibration, and empirical study on its impact on organizational creativity and firm performance, Inf. Manag. 58 (2021) 103434. https://doi.org/10.1016/j.im.2021.103434

[37] M.A. Onari, M.J. Rezaee, M. Saberi, M.S. Nobile, An explainable data-driven decision support framework for strategic customer development, Knowl.-Based Syst. 295 (2024) 111761. https://doi.org/10.1016/j.knosys.2024.111761

[38] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, et al., Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI, Inf. Fusion 58 (2020) 82–115. https://doi.org/10.1016/j.inffus.2019.12.012

[39] A. Adadi, M. Berrada, Peeking inside the black-box: A survey on explainable artificial intelligence (XAI), IEEE Access 6 (2018) 52138–52160. https://doi.org/10.1109/ACCESS.2018.2870052

[40] A. Artelt, R. Visser, B. Hammer, "I do not know! but why?" — Local model-agnostic example-based explanations of reject, Neurocomputing 558 (2023) 126722. https://doi.org/10.1016/j.neucom.2023.126722

[41] J.D.C. Little, A proof for the queuing formula: L = λW, Oper. Res. 9 (1961) 383–387. https://doi.org/10.1287/opre.9.3.383

[42] S. Lessmann, B. Baesens, H.V. Seow, L.C. Thomas, Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research, Eur. J. Oper. Res. 247 (2015) 124–136. https://doi.org/10.1016/j.ejor.2015.05.030

[43] [dataset] H. Hofmann, Statlog (German Credit Data), UCI Machine Learning Repository (1994). https://doi.org/10.24432/C5NC77

[44] [dataset] R. Quinlan, Statlog (Australian Credit Approval), UCI Machine Learning Repository (1987). https://doi.org/10.24432/C59012

[45] [dataset] I. Yeh, Default of Credit Card Clients, UCI Machine Learning Repository (2009). https://doi.org/10.24432/C55S3H

[46] [dataset] B. Baesens, D. Rösch, H. Scheule, Credit Risk Analytics: Measurement Techniques, Applications, and Examples in SAS, Wiley, Hoboken, NJ, 2016 (HMEQ data set).

[47] C. Nadeau, Y. Bengio, Inference for the generalization error, Mach. Learn. 52 (2003) 239–281. https://doi.org/10.1023/A:1024068626366

[48] S. Holm, A simple sequentially rejective multiple test procedure, Scand. J. Stat. 6 (1979) 65–70.